LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Geometrical characteristics of aerofoil, types of aerofoil, definition of terms
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Geometrical characteristics of aerofoil, types of aerofoil, definition of terms.
5. Specific Objectives	6.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Examination of aerofoil cross-section models; Safety: Handle sharp edges with care; Supervision: Instructor monitors model handling.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an aerofoil?
(b) Define the term 'lift' in the context of flight.
(c) List three factors affecting aerodynamic performance.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of aerofoil • Leading and trailing edges • Chord line and chord length • Camber and camber line • Mean aerodynamic chord	L & Ds	PPT, IB	15 min
(b)	• Symmetrical aerofoil types • Asymmetrical aerofoil types • Thick vs thin aerofoils • High lift vs low drag profiles • Comparison of geometrical features	L & Ds	PPT, PDF	20 min
(c)	• Definition of angle of attack • Stall angle and critical angle • Zero lift angle definition • Pressure distribution concepts • Flow separation points	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a symmetrical and asymmetrical aerofoil.
(b) Describe how the camber line influences lift generation.
(c) Define the stall angle and its significance in flight.

12. Home Assignment and Follow Up	Prepare a brief note on how the geometrical characteristics of an aerofoil influence the lift coefficient, linking this to the principles of flow over bodies covered in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Bernoulli's theorem for compressible and incompressible Flow. Equation of continuity, Lift generation on aerofoil. Reynolds number.	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min